

CHECKLIST

SMALL CRAFT - ELECTRICAL SYSTEMS - ALTERNATING AND DIRECT CURRENT INSTALLATIONS

Ref.: EN ISO 13297:2021 + A1:2022+A11:2023 (ISO 13297:2020)

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	



Subject to check		Requirements	Checked ?
The metallic hull of a craft must not be used as a circuit conductor	4.1	[Yes / NA ?]	
Craft with both DC and AC electrical systems must distribute power through either separate panel boards, a common panel board with a partition, or another effective method that clearly separates the AC and DC sections. These sections must also be clearly labeled.	4.2	[Yes / NA ?]	
Switches and controls must be labeled to indicate their function unless their purpose is obvious	4.4	[Yes / NA ?]	
The system is fully insulated two-wire or two-wire with negative ground.	5.1	[Yes / NA ?]	
If d.c. system with negative ground, main ground/earthing is the negative terminal or main ground bus.	5.2	[Yes / NA ?]	
Multiple battery banks have a common negative connection, if not excepted as dedicated isolated system, e.g. electric propulsion system.	5.1	[Yes / NA ?]	
Protective devices are provided at the source of power	5.3	[Yes / NA ?]	
The protective conductor insulation is green or green with yellow stripe in A.C. systems. Neither colour is used for current carrying conductors. Note: the equipotential conductor of the D.C. system also uses green or green with a yellow stripe insulation.	6.1	[Yes / NA ?]	
Fully insulated D.C. systems: A.C. protective conductor is connected to metallic hull respectively to the craft external ground/earth for a non-conductive hull.	6.2	[Yes / NA ?]	
The a.c. protective conductor has a single final connection to metallic hull respectively to the main grounding point/earthing point. connection	6.3	[Yes / NA ?]	
Metallic craft: connection of the protective conductor is above any anticipated water accumulation.	6.4	[Yes / NA ?]	
Metallic housings or enclosures of appliances are connected to the protective conductor.	6.5	[Yes / NA ?]	
Individual circuits cannot be energized by more than one source at a time.	6.6	[Yes / NA ?]	
The transfer and cut-off from one powering source to another fullfills 6.6	6.6	[Yes / NA ?]	
Energized parts are IP2X (IEC 60529) or guarded by other protective means against accidental contact. Access only by use of tools if not IP 2X protected.	6.7	[Yes / NA ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:		
Watercraft model name:		
A suitable warning sign shall be displayed in way of Energised parts	6.7,7.2	[Yes / NA ?]
 General warning sign ISO 7010 — W001  Warning: Electricity ISO 7010 — W012  Warning: Flammable material ISO 7010 — W021  Refer to instruction manual/booklet ISO 7010 — M002 a) Suggested warning sign using symbols WARNING — To minimize shock and fire hazards: 1 Turn off craft's shore power connection switch before connecting or disconnecting shore power cable. 2 Connect shore power cable to craft's inlet before connecting to shore power source. 3 If incorrect polarity is indicated, immediately disconnect cable. 4 Disconnect shore power cable at shore power source first. 5 Close shore power inlet cover tightly. DO NOT ALTER SHORE POWER CABLE CONNECTORS		
The neutral conductor is only grounded at the source of power.	6.8	[Yes / NA ?]
The shore power neutral is grounded through the shore power cable and not grounded on board of the craft (see exceptions).	6.8	[Yes / NA ?]
If fitted in the protective conductor, the galvanic isolator is fail-safe.	6.9	[Yes / NA ?]
If the polarity of the system has to be maintained for the proper operation, a reverse polarity indicating devices fitted.	6.10	[Yes / NA ?]
reverse polarity device is not necessary if boat uses an un-polarized device or an isolation transformer	6.10	[Yes / NA ?]
Label: Shore power inlets marked to indicate voltage and currents.	7.1	[Yes / NA ?]
 Warning: Electricity ISO 7010 — W012  Refer to instruction manual/booklet ISO 7010 — M002 Affixed: in the vicinity of the shore power inlet.		

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:			
Watercraft model name:			
Label: A permanently mounted waterproof warning sign for energized parts / or in language acceptable in the country of use.		7.2	[Yes / NA ?]
 General warning sign ISO 7010 — W001  Warning: Electricity ISO 7010 — W012  Warning: Flammable material ISO 7010 — W021  Refer to instruction manual/booklet ISO 7010 — M002 a) Suggested warning sign using symbols WARNING — To minimize shock and fire hazards: 1 Turn off craft's shore power connection switch before connecting or disconnecting shore power cable. 2 Connect shore power cable to craft's inlet before connecting to shore power source. 3 If incorrect polarity is indicated, immediately disconnect cable. 4 Disconnect shore power cable at shore power source first. 5 Close shore power inlet cover tightly. DO NOT ALTER SHORE POWER CABLE CONNECTORS			
Affixed: At the panel-board.			
Batteries are installed in dry and vented location above bilge water level.	8.1	[Yes / NA ?]	
Batteries are protected against mechanical damage by location or enclosure.	8.5	[Yes / NA ?]	
Batteries installed above or below a fuel tank or fuel filter shall be isolated from fuel components	8.6	[Yes / NA ?]	
Metallic components of the fuel system within 300 mm above battery top are electrically insulated.	8.7	[Yes / NA ?]	
Connected battery cable terminals do not depend on spring tension.	8.8	[Yes / NA ?]	
Battery disconnect switch in the positive conductor if earthed negative system.	9.1	[Yes / NA ?]	
Battery disconnect switch in the positive and negative conductor if fully insulated two-wire d.c. system.	9.1	[Yes / NA ?]	
Battery disconnect switch can be reached quickly and safely without use of tools, positioned as close as practical to the battery / group of batteries. See exceptions.	9.1	[Yes / NA ?]	
Inverter(s) is/are installed as per 11.1	11.1	[Yes / NA ?]	
Inverter outlet circuits are protected.	11.2	[Yes / NA ?]	
If installed in conditions according to clause 25, inverter(s) shall have IGP and be marked accordingly.	11.3	[Yes / NA ?]	
Label: d.c. conductor terminals or conductor label d.c. +/- or POS/NEG or +/-.	11.4	[Yes / NA ?]	
Affixed: on conductor terminal or conductor.			
Label: The inverter provides ready connection of three or more conductors with connectors, eg. L1, G, N.	11.5	[Yes / NA ?]	
Affixed: on the terminals or conductors.			
The inverter integral switch is switching all live conductors and disconnect grounded conductor when used with an external supply	11.7	[Yes / NA ?]	
Label: If an inverter is installed:	11.8	[Yes / NA ?]	
WARNING — ELECTRICAL SHOCK HAZARD To avoid electrical shock hazard: disconnect AC shore power and DC battery power to inverter before opening panel			
Affixed: all panels containing AC live connections			

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:			
Watercraft model name:			
Inverter(s)/charger(s) is/are marked/identified with: -input voltage, amperage and frequency; -nominal output voltage and frequency; -d.c. output voltage at float or shut off, if applicable; -continuous output current at 12 V at specified input voltage at 25°C; -battery type; -warning concerning internal charged capacitor hazard when servicing.	11.9	[Yes / NA ?]	
Inverter(s) is/are marked/identified with: -input voltage and amperage; -continuous output current at rated voltage; -nominal output voltage and frequency; -surge capability and duration.	11.10	[Yes / NA ?]	
An overcurrent protection is installed at a maximum of 175mm of every power source	12.1	[Yes / NA ?]	
If the battery conductor is contained in a sheath or enclosure, the overcurrent protection may be placed up to 1,8 m from the battery.	12.1	[Yes / NA ?]	
If the conductor is connected to a source of power other than a battery terminal, and contained in a sheath or enclosure, the overcurrent protection may be placed up to 1 m from the power source. See exemption for self-limited devices.	12.1	[Yes / NA ?]	
Voltage rating of fuses and circuit breakers are not less than the nominal circuit voltage.	12.2	[Yes / NA ?]	
Ungrounded conductors of chargers are equipped with overcurrent protection at a maximum distance of 1.8m from the source	12.3	[Yes / NA ?]	
Double pole circuit breakers that open both live and neutral conductors are installed in the unpolarized system.	13.1.1	[Yes / NA ?]	
No fuses are installed in the unpolarized system. In polarized system a fuse shall interrupt the active (phase) conductor.	13.1.2	[Yes / NA ?]	
Overcurrent protection device does not exceed the maximum admissible current of the conductor on which it operates. See table A.1	13.1.5	[Yes / NA ?]	
Double pole circuit breakers that open both live and neutral conductors are installed in all supply circuits	13.2.1	[Yes / NA ?]	
A manually reset trip-free circuit-breaker is installed within 0,5 m of the source of power or, the conductor from the source of power to the panel-board circuit-breaker is contained within a protective covering or, within a conduit or cable trunking or equivalent protective covering.	13.2.2	[Yes / NA ?]	
Main shore power inlet circuit breaker: additional fuses or circuit-breakers are provided within 3 m of the inlet or attachment point to the electrical system in the craft, measured along the conductor for those located over 3 m from the shore-power inlet connection or the electrical attachment point of a permanently installed shore-power cord.	13.2.2	[Yes / NA ?]	
Overcurrent protection is provided for isolation, single or coupled transformers. The maximum overcurrent rating of the protection shall not be higher than 125% of the primary current.	13.2.3	[Yes / NA ?]	



Checklist_Evaluation_Module B_G en260703



Watercraft manufacturer:		
Watercraft model name:		
Overcurrent protection is provided for live conductors of each branch circuit of a polarised system at the point of connection to the main panel board bus.	13.3.1	[Yes / NA ?]
Branch circuit in unpolarised systems: both conductors are provided with overcurrent protection by double-pole circuit-breakers and double-pole switches, if used, at the point of connection to the main panel-board bus.	13.3.2	[Yes / NA ?]
Craft earth leakage protection provided in all sources by one or more double-pole RCDs (30mA trip sensitivity & 100 ms max. trip time).	14.2	[Yes / NA ?]
The RCD device has an internal circuit for manual testing of the trip function.	14.3	[Yes / NA ?]
The front side of the panel board is readily accessible, rear side accessible.	15.1	[Yes / NA ?]
Label: The panel board is permanently marked with the system voltage and frequency. Affixed: On the panel board.	15.2	[Yes / NA ?]
Switches and controls that operates the craft are obvious	15.3	[Yes / NA ?]
AC panel board is provided with an indication of the on/off system status	16.1	[Yes / NA ?]
When designed to supply motor circuits or a generator is installed, a system voltmeter is installed on the panel board.	16.2	[Yes / NA ?]
A visible means is provided on the panel board, indicating the inverter is active on line and/or standby (voltmeter or lamp).	16.3	[Yes / NA ?]
Label: If an inverter is installed: WARNING — ELECTRICAL SHOCK HAZARD Craft is equipped with a DC to AC power inverter. To avoid serious injury or death from electrical shock: disconnect AC shore power and DC battery power to inverter before opening panel or servicing electrical systems. Affixed: on the panel board when an inverter is installed.	16.4	[Yes / NA ?]
Neutral conductors are white or light blue.	19.4	[Yes / NA ?]
Protective conductors are green or green/yellow.	19.4	[Yes / NA ?]
Neutral conductors or active phase of a AC system are not yellow, green or green with a yellow stripe	19.4	[Yes / NA ?]
Conductor connections located protected from the weather are minimum IP 55.	20.1	[Yes / NA ?]
Connections above deck exposed to intermittent immersion are IP 67.	20.1	[Yes / NA ?]
Conductors are supported throughout their length in conduits, cable trunking or trays, or by individual supports at maximum intervals of 450 mm.	20.2	[Yes / NA ?]
Conductors exposed to physical damage are protected by sheaths, conduits or other means. Bulkhead and structural members passages for conductors are protected against chafing.	20.3	[Yes / NA ?]
The d.c. circuit is separated from the a.c. circuit by an earthed metal screen in a multicore cable.	20.4	[Yes / NA ?]
The d.c. circuit is separated from the a.c. circuit by separated conduits or trunking systems.	20.4	[Yes / NA ?]
The d.c. circuit is separated from the a.c. circuit by physical partition	20.4	[Yes / NA ?]

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:			
Watercraft model name:			
The d.c. circuit is separated from the a.c. circuit by more than 100mm, when fixed directly to a surface	20.4	[Yes / NA ?]	
Conductors are routed above bilge water level or at least 25mm above automatic bilge pump switch.	20.5	[Yes / NA ?]	
No Conductors connections below the bilge pump switch, if not enclosed in an IP67 enclosure	20.5	[Yes / NA ?]	
Conductors are routed 50 mm away from water-cooled exhaust components, unless an equivalent thermal barrier is provided.	20.6	[Yes / NA ?]	
Conductors are routed 250 mm away from dry exhaust components, unless a thermal barrier is provided.	20.6	[Yes / NA ?]	
Electrical contact is obtained without damaging the conductors also assuring mechanical linkage, by the use of ring or othe captive spade type fittings	20.9	[Yes / NA ?]	
No twist-on connectors (wire nuts) are used.	20.10	[Yes / NA ?]	
Exposed shanks of terminals are protected against accidental shorting. Those for the grounding systems may be exempt.	20.11	[Yes / NA ?]	
Solderless crimp-on terminals and connectors are attached with a suitable crimping tool.	20.12	[Yes / NA ?]	
Tensile values for connectors are in compliance with Table 1.	20.13	[Yes / NA ?]	
No more than four conductor terminals shall be secured to one stud	20.14	[Yes / NA ?]	
Sheathed conductors and battery conductors to the battery disconnect switch shall be supported at maximum intervals of 300 mm with the first support not more than 1 m from the terminal	21	[Yes / NA ?]	
The a.c. and d.c. system sockets are not interchangeable.	22.1	[Yes / NA ?]	
Socket outlets installed in areas subject to flooding or momentary submersion are in IP 55enclosures , as a minimum, and with or without the plug	22.2	[Yes / NA ?]	
Socket outlets installed in areas subject to flooding or momentary submersion are in IP 67 enclosures , as a minimum, and with or without the plug	22.3	[Yes / NA ?]	
Shore power inlets have minimum rating of IP44 when mated with plug.	23.1	[Yes / NA ?]	
Socket outlets have a grounding/earthing terminal	23.2	[Yes / NA ?]	
Socket outlets for the galley area are located such that chords can be plugged in without crossing above a stove or sink.	23.3	[Yes / NA ?]	
Appliances and fixed AC equipment have exposed conductive parts connected to the craft protective conductor, unless double-insulated construction	24.1	[Yes / NA ?]	
The following questions shall be filled in by the watercraft manufacturer and appropriate documentation shall be			
Subject to check		Requirements	Checked ?
Wiring diagrams detailing circuits, components, and conductors must be provided	4.3	[Yes ?]	
AC installation is tested according to Annex C	4.3	[Yes / NA ?]	
Voltage ranges of d.c. equipment functions within voltage range (75% - 133%) at battery terminals; exception for equipment requiring a higher minimum fulfilled requirement of Annex A	5.4	[Yes / NA ?]	
Calculated voltage drop does not exceed 10% of nominal voltage for conductors.	5.5	[Yes / NA ?]	

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:			
Watercraft model name:			
The value of current flow for overcurrent protection devices for motor loads is consistent with the demand load characteristics of the protection circuit.	5.6	[Yes / NA ?]	
Correct voltage drop ensure functionality of safety-critical equipment	5.6	[Yes / NA ?]	
Installed batteries are capable of inclinations of 30° without leakage.	8.3	[Yes / NA ?]	
Monohull sailing craft: spilled electrolyte is contained up to 45° without leakage.	8.3	[Yes / NA ?]	
Movement of batteries is less than 10 mm at force twice the battery weight.	8.4	[Yes / NA ?]	
30-second rating of a battery switch shall be capable of operating a maximum current equal to the maximum current in cranking motor and a minimum current equal to the lesser of the switch cable maximum current or the maximum rating current of the protection device	9.2	[Yes / NA ?]	
Remote controlled battery is doubled by a manual command	9.3	[Yes / NA ?]	
AC installation is powered by single shore power cable whose section is able to carry the required current of the system	10.1	[Yes / NA ?]	
AC installation is powered by several shore power cables whose sections are able to carry the required current of the system	10.1	[Yes / NA ?]	
AC installation is powered by an inverter	10.1	[Yes / NA ?]	
AC installation is powered by AC generator(s)	10.1	[Yes / NA ?]	
AC installation is powered by a combination of shore power cables, inverter or inverter charges, providing that each source is isolated from the other sources	10.1	[Yes / NA ?]	
AC generator power feeder is protected from overcurrent at a maximum of 120% output rating, if AC generator is not equipped with an overload protection which does not exceed the maximum output rating	10.3	[Yes / NA ?]	
Each AC motor is protected from overcurrent or thermal runout, whenever the motor is not ment to handle locked shaft	13.1.4	[Yes / NA ?]	
RCD are trip-free type	14.1	[Yes / NA ?]	
In engine spaces the conductor insulation is temperature rated at $\geq 70^{\circ}\text{C}$ and insulation is oil-resistant or be protected by conduit or sleeving. Current carrying capacity is derated to Annex A.	17.1	[Yes / NA ?]	
If conductors are longer than 175mm are at least 1 mm^2	17.2	[Yes / NA ?]	
Sheathed conductors are at least $0,75\text{mm}^2$ and can extend un sheathed to a maximum distance of 800mm	17.2	[Yes / NA ?]	
DC conductors are in copper and isolated by a fire resistant material	18.1	[Yes / NA ?]	
DC conductors are sized according to Table 1. or according to the conductor manufacturer current rating	18.2	[Yes / NA ?]	
The protective conductor insulation is green or green with yellow stripe. Neither colour is used for current carrying conductors. Note: the equipotential conductor of the D.C. system also uses green or green with a yellow stripe insulation.	18.3	[Yes / NA ?]	
Other colors used to identify DC positive conductor are provided in a wiring diagram	18.4	[Yes / NA ?]	
Black and yellow cable insulations are only used for DC positive conductors.	18.5	[Yes / NA ?]	
If case where an AC system onboard uses the black insulation for live conductors, yellow insulation is used for DC negative	18.5	[Yes / NA ?]	



compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:			
Watercraft model name:			
Other colors used to identify DC negative conductor are provided in a wiring diagram	18.6	[Yes / NA ?]	
brown, white or light blue insulation colour in the are not used in DC system, or are clearly separated from AC conductors and identified as DC	18.6	[Yes / NA ?]	
Conductors and flexible cords have a minimum rating of 300/500 V.	19.1	[Yes / NA ?]	
Conductor insulation outside engine spaces is temperature rated at $\geq 60^{\circ}\text{C}$.	19.2	[Yes / NA ?]	
Protective conductor cross sections are compliant with the 19.3	19.2	[Yes / NA ?]	
Studs, nuts and washers are corrosion resistant and galvanically compatible.	20.7	[Yes / NA ?]	
No bares wires or screw connections	20.8	[Yes / NA ?]	
Friction type connectors used only in circuits not exceeding 20 A and with separation force $> 20\text{N}$.	20.9	[Yes / NA ?]	
Receptacles/sockets have a voltage rating matching the power sources.	23.4	[Yes / NA ?]	
Appliances and AC equipments are equipped with overcurrent protection	24.1	[Yes / NA ?]	
Electrical components and devices installed in compartments which can contain explosive vapour and gases are ignition-protected in accordance with ISO 8846:1990	25.1	[Yes / NA ?]	
Instructions/Warnings to be included in the owner's manual			
94 Wiring diagram with conductor identification.	Annex B	[Yes ?]	
95 Do not modify the craft's electrical system or relevant drawings. Installation, alterations and maintenance should be performed by a competent marine electrical technician. Inspect system at least biennially.	Annex B	[Yes ?]	
96 Disconnect shore power connections when system is not in use.	Annex B	[Yes ?]	
97 Connect metallic housings or enclosures of installed electrical appliances to the protective conductor system in the craft (green or green with a yellow stripe conductor).	Annex B	[Yes ?]	
98 Use double-insulated or grounded (earthed) electrical appliances.	Annex B	[Yes ?]	
99 If reverse polarity indicator is activated, do not use electrical system. Correct polarity fault before activating the electrical system on the craft.	Annex B	[Yes / NA ?]	
100 Do not alter shore power cable connectors, use only compatible cable connectors and shore power receptacles.	Annex B	[Yes ?]	
101 Instructions on monthly testing of the RCD.	Annex B	[Yes ?]	
102 WARNING — Do not allow shore power cable end to hang in the water. An electrical field can result which can cause injury or death to nearby swimmers.	Annex B	[Yes ?]	



Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:						
Watercraft model name:						
103	WARNING — To minimize shock and fire hazards: -Turn off craft's shore power connection switch before connecting or disconnecting shore power cable. -Connect shore power cable to craft's inlet before connecting to shore power source. -If reverse polarity indicator is activated, turn off craft's shore power connection switch immediately (if fitted). -Disconnect shore power cable at shore power source first. -Close shore power inlet cover tightly.	Annex B	[Yes ?]			
Comments: <table border="1" style="width: 100%; height: 40px;"> <tr><td></td></tr> <tr><td></td></tr> <tr><td></td></tr> </table>						